Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

High-Performance Computing Lab for CSE **2026**

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Report for Project 1

Due date: 09 March 2026, 18:00 (Europe/Zurich)

HPC Lab for CSE 2025 — Submission Instructions

Following these instructions is mandatory: Submissions that do not comply **will not be considered**.

- **Submission Platform**: Assignments must be submitted via SAM-UP.
- **Required Files**: Your submission must include:
  - **Report**: A PDF summarizing your results and observations, based on this LaTeX template.
    - File name: `project_number_lastname_firstname.pdf`
    - You are allowed to discuss all questions with anyone you like; however: (i) your submission must list anyone you discussed problems with and (ii) you must write up your submission independently.
  - **Source Code Archive**: A **compressed tar archive**, named `project_number_lastname_firstname.tar.gz`, containing:
    - All source code files.
    - Build files and scripts (e.g., `Makefile`, batch job scripts, plot scripts, etc). If you modified any provided files, ensure they still work as expected.
    - The full LaTeX source of your report including all needed files to build the report (figures, listings, etc).
  - **Size Limit**: The total submission (PDF + archive) must not exceed **25 MB**.
  - **File Naming Rules**: Use only `ASCII` characters (avoid spaces, special symbols, and non-English letters).
- **Grading Process**: The TAs will evaluate your project based on:
  - Your report write-up.
  - Your code implementation.
  - Benchmarking your code's performance.

1. Euler warm-up [10 points]

Answer to Question 1.1

Answer to Question 1.2

Answer to Question 1.3

2. Performance characteristics [50 points]

2.1. Peak performance [16 Points]

Proceeding as in the example given in the project description, the hardware specifications can be obtained from the Euler webpage.
For Euler VII (phase 1), the system consists of $n_{\text{nodes}} = 292$ nodes, each equipped with $n_{\text{sockets}} = 2$ sockets and $n_{\text{cores}} = 64$ cores per socket, operating at a frequency $f = 2.6\,\text{GHz}$.
For Euler VII (phase 2), the system consists of $n_{\text{nodes}} = 248$ nodes, each with $n_{\text{sockets}} = 2$ sockets and $n_{\text{cores}} = 64$ cores per socket, operating at a frequency $f = 2.45\,\text{GHz}$.

Using the uops.info database provided, we determine both the superscalability factor and the FMA factor for the corresponding AMD architectures.
For both Euler VII phases, which are based on AMD Zen2 and AMD Zen3 respectively, we obtain $n_{\text{super}} = 4$ and $n_{\text{FMA}} = 2$.

The FMA factor is determined by observing that all instructions marked as **FMA** in the table have a corresponding number of micro-operations equal to one. This indicates that the fused multiply-add is executed as a single micro-operation by dedicated hardware. Since an FMA instruction performs the operation $a \cdot b + c$, it consists of one floating-point multiplication and one floating-point addition. Therefore, each FMA instruction corresponds to two floating-point operations, and thus $n_{\text{FMA}} = 2$.

The superscalability is defined as the number of floating-point operations that the FPU of a core can execute in parallel within a single clock cycle. To determine this value, we analyze the instruction `VFMADD132PD (YMM, YMM, YMM)` reported in the uops.info table. The table specifies a throughput of 0.50 cycles per instruction, one micro-operation, and execution on ports FP0/1. A throughput of 0.50 cycles per instruction implies that

$$\frac{1}{0.50} = 2$$

FMA instructions can be issued per cycle. The port information FP0/1 confirms the presence of two independent floating-point execution pipelines capable of executing FMA instructions in parallel.

Since two FMA instructions can be executed per cycle and each FMA instruction corresponds to two floating-point operations, the total number of floating-point operations per cycle is

$$2 \times 2 = 4.$$

Hence, for both AMD Zen 2 and AMD Zen 3, the superscalability is

4 floating-point operations per cycle per core.

To determine n_{SIMD}, we inspect the floating-point instructions reported in the uops.info table, where instructions such as `VFMADD132PD (YMM, YMM, YMM)` appear. The presence of YMM operands indicates the use of 256-bit AVX registers, since YMM registers are defined as 256-bit wide in the AVX/AVX2 instruction set. As both AMD Zen 2 and Zen 3 execute these YMM-based instructions, they necessarily support AVX2 with 256-bit SIMD registers. A double-precision floating-point number occupies 64 bits; therefore, the number of doubles that can be processed concurrently by one SIMD instruction is

$$n_{\text{SIMD}} = \frac{256}{64} = 4.$$

Hence, each SIMD instruction can operate on four double-precision values in parallel.

Parameter	**Euler VII (Phase 1)**	**Euler VII (Phase 2)**
n_{nodes}	292	248
n_{sockets}	2	2
n_{cores} (per socket)	64	64
Clock frequency f	2.6 GHz	2.45 GHz
n_{FMA}	2	2
n_{super} (FP ops/cycle/core)	4	4
n_{SIMD} (doubles per SIMD instr.)	4	4

Table 1: Summary of hardware parameters and architectural factors used for peak performance estimation.

We can now compute all the peak performance parameters for both Euler VII phases :

$$P_{\text{core},1} = n_{\text{super}} \times n_{\text{FMA}} \times n_{\text{SIMD}} \times f = 4 \times 2 \times 4 \times 2.6\,\text{GHz} = 83.2\,\text{GFLOPS/core}.$$
$$P_{\text{CPU},1} = n_{\text{cores}} \times P_{\text{core},1} = 64 \times 83.2\,\text{GFLOPS/core} = 5.3248\,\text{TFLOPS/CPU}.$$
$$P_{\text{node},1} = n_{\text{sockets}} \times P_{\text{CPU},1} = 2 \times 5.3248\,\text{TFLOPS/node} = 10.6496\,\text{TFLOPS/node}.$$
$$P_{\text{cluster},1} = n_{\text{nodes}} \times P_{\text{node},1} = 292 \times 10.6496\,\text{TFLOPS/node} = 3110.08\,\text{TFLOPS}.$$

$$P_{\text{core},2} = n_{\text{super}} \times n_{\text{FMA}} \times n_{\text{SIMD}} \times f = 4 \times 2 \times 4 \times 2.45\,\text{GHz} = 78.4\,\text{GFLOPS/core}.$$
$$P_{\text{CPU},2} = n_{\text{cores}} \times P_{\text{core},2} = 64 \times 78.4\,\text{GFLOPS/core} = 5.017\,\text{TFLOPS/CPU}.$$
$$P_{\text{node},2} = n_{\text{sockets}} \times P_{\text{CPU},2} = 2 \times 5.017\,\text{TFLOPS/node} = 10.034\,\text{TFLOPS/node}.$$
$$P_{\text{cluster},2} = n_{\text{nodes}} \times P_{\text{node},2} = 248 \times 10.034\,\text{TFLOPS/node} = 2488.432\,\text{TFLOPS}.$$

2.2. Memory Hierarchies [8 Points]

2.2.1. Cache and main memory size

2.3. Bandwidth: STREAM benchmark [10 Points]

2.4. Performance model: A simple roofline model [16 Points]

3. Auto-vectorization [10 points]

4. Matrix multiplication optimization [30 points]